

The Pop Bottle Magnetometer – Background Information and Additional Resources

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Background information:

- The word magnetometer is a combination of the words **magnet + meter**.
- A **magnetometer** is any device capable of measuring the strength, direction or changes in Earth's magnetic field. A field compass is a simple magnetometer that gives the direction of Earth's magnetic field (Ungvarsky, 2023).
- The first magnetometer, built by Carl Friedrich **Gauss** in 1832, was a simple instrument that consisted of a bar magnet suspended horizontally by a fibre. He used this to test the strength and location of the Earth's magnetic field. The unit of magnetic field strength was termed 'Gauss' in his honour (Ungvarsky, 2023).
- Nikola **Tesla** built on Gauss' work and created several devices involving magnetometers. 'Tesla' is also a commonly used unit for magnetic measurements (Ungvarsky, 2023).
- Early magnetic instruments were used on the 1873-1876 **HMS Challenger expedition**, and the measurements collected greatly contributed to the study of Earth's magnetism (Creak, 1889). Instruments used included a Fox circle, Barrow's circle, Unifilar Magnetometer, Kater's azimuth compass and a small vibrating needle (Tizard, 1882).
- Sir Douglas **Mawson** used magnetometers in his Australasian Antarctica Expedition 1911-1914. Mawson's men were measuring the strength and direction of Earth's magnetic field. This research had great commercial and military significance (Henderson, 2015).

You can find an example of one of Mawson's magnetometers at the [Mawson's Huts Replica Museum](#) in Hobart, Tasmania (see **Figure 1**). The instrument on display at the museum is very similar to the pop bottle magnetometer in this project. It contains a

magnet that you can move by pulling a string. You then watch a compass needle and a laser spot on a recording drum move in response.

If you are based in/ visiting Hobart, why not see if you can find this magnetometer in the museum? The museum is located on the waterfront on the corner of Argyle and Morrison Streets.



Figure 1: The interactive magnetometer on display at the [Mawson's Huts Replica Museum](#) in Hobart (Mawson's Huts Replica Museum, Hobart, Tasmania).

- The Earth has a magnetic field which extends out into space and forms the **magnetosphere** (Olsen & Pauluhn, 2019) (see **Figure 2**).
- A **geomagnetic storm** occurs when solar particles emitted from the Sun (the solar wind) interact with Earth's magnetosphere (SWS BoM).
- Geomagnetic storms compress Earth's magnetosphere and can cause **angular shifts** in the location of magnetic north. These shifts can be recorded by a pop bottle magnetometer. Most geomagnetic storms produce shifts between 0.5° and 1° (NASA, 2015).

- Earth's magnetosphere also undergoes a **daily cycle** where the sunward side is compressed, and night-time side is extended. This regular cycle can also be recorded by a pop bottle magnetometer, with around 0.5° movement expected daily (AuroraWatchUK).

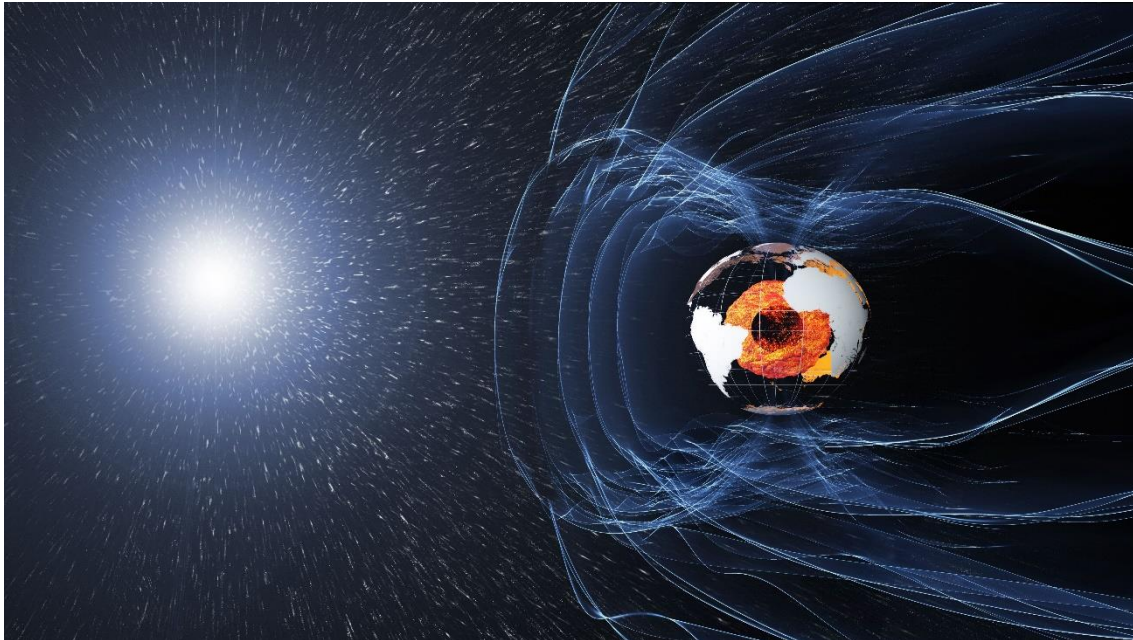


Figure 2: The Earth's magnetosphere acts as a protective force field and shields our planet from harmful solar particles ([ESA/ATG medialab](#))

In this project, students will construct a simple magnetometer similar to that used by Gauss and Mawson and understand how it can be used to measure changes in Earth's magnetic field.

References

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Bureau of Meteorology Australian Space Weather Forecasting Centre (SWS BoM). *Aurora Photos and the Aurora Explained*. Available at: <https://www.sws.bom.gov.au/Educational/1/1/4>

Creak, E. (1889). Report on the Magnetical Results of the Voyage of H.M.S. "Challenger." *Nature*, 41 (105–108). <https://doi.org/10.1038/041105d0>

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NASA (2015). *Soda Bottle Magnetometers for Space Weather Studies*. Available at: <https://science.nasa.gov/learn/heat/resource/soda-bottle-magnetometers-student-guide/>

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Ungvarsky, J. (2023). Magnetometer <https://www.ebsco.com/research-starters/history/magnetometer>

Additional resources:

Additional classroom activities using pop bottle magnetometers, including instructions for more advanced designs: NASA (2015). *Soda Bottle Magnetometers for Space Weather Studies*. Available at:

<https://science.nasa.gov/learn/heat/resource/soda-bottle-magnetometers-student-guide/>

For information on how Earth's magnetic field is generated and how the solar wind influences Earth's magnetosphere: Olsen, N. and Pauluhn, A. (2019). Exploring Earth's magnetic field – Three make a Swarm. *Spatium*, 2019(43), pp. 3-15. Available at:
https://backend.orbit.dtu.dk/ws/portalfiles/portal/185293289/Exploring_Earths_magnetic_field.pdf

A short ESA animation showing Earth's magnetosphere shielding the planet from the solar wind: European Space Agency (ESA) (2012). *The force that protects our planet* [Video]. ESA Multimedia. Available at:
https://www.esa.int/ESA_Multimedia/Videos/2012/09/The_force_that_protects_our_planet